



INDUSTRIAL AUTOMATION

PLC, NETWORK COMMUNICATION & MOTOR CONDITION MONITORING

A Practical Internship Study and Node-RED Monitoring Prototype

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ACKNOWLEDGEMENT

I express my sincere gratitude to Tata Steel and the concerned industrial team for providing the opportunity to study industrial automation, Programmable Logic Controllers (PLCs), network communication and maintenance practices in an industrial environment. The exposure helped connect classroom concepts with real control-system hardware, field instrumentation and operator interfaces. I am grateful to my industrial mentors and supervisors for their guidance, practical explanations and support during the internship. Their inputs helped me understand not only how PLC systems operate, but also how engineers approach reliability, fault diagnosis, communication and maintenance.

I also acknowledge the reference material used during the study, including the PLC and networking report and the troubleshooting presentation supplied for this work. These materials provided the technical foundation for the sections on PLC architecture, redundant power, industrial connections, the Siemens IM 153-1 interface, I/O troubleshooting and motor-drive concepts.

Finally, I thank my academic department, faculty members, colleagues and everyone who contributed directly or indirectly to the completion of this internship project.

ABSTRACT

Industrial production depends on reliable automation systems in which PLCs, sensors, drives, communication networks and operator interfaces work together. This report combines the practical learning of PLC architecture and industrial networking with the development of a supervisory motor condition-monitoring prototype using MQTT and Node-RED.

The first part of the work establishes the industrial foundation: PLC CPU, input/output modules, communication interfaces, HMI, hardware configuration, redundant power concepts, ring and star networking, industrial cabling, Siemens IM 153-1 remote-I/O interfacing, troubleshooting of digital and analog signals, and DC-drive fundamentals.

The second part converts these concepts into an applied monitoring workflow for a motor. The prototype receives motor data through the MQTT topic "industrial/motor", processes temperature, vibration, current, motor and alarm states in Node-RED, calculates a rule-based health score, generates threshold-based alerts, counts alarms and presents the information on a web dashboard. The dashboard contains live gauges, motor/alarm status, health score, trend charts, alert indicators and a last-update timestamp.

The PLC is the primary industrial control system, while my Node-RED application acts as a supervisory monitoring layer. In a real implementation, sensors would provide temperature, vibration and current information to the PLC through its I/O system. Selected process data could then be transferred through an approved industrial gateway to an MQTT-based monitoring layer. Node-RED would process that data, calculate a condition score, generate supervisory alarms and display real-time trends. The PLC would continue to handle deterministic control and interlocking, while Node-RED would support visualization and maintenance decisions.

A future deployment can connect actual PLC analog/digital signals or industrial gateways to the same monitoring architecture, add historian storage and introduce machine-learning models for failure prediction.

Key outcome

The report links field-level instrumentation → PLC/I/O → industrial networking → data gateway → Node-RED analytics → operator dashboard → maintenance decision.

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1. INTRODUCTION AND INDUSTRIAL CONTEXT

1.1 Introduction to Wire Rolling Mill

This steel division of TATA STEEL at Jamshedpur is one of the largest specialty steel manufacturers in India. This division produces over 300,000mt presently. It serves a host of industries like Automobile, Auto ancillaries, General engineering, Fasteners, Railways, Defence and Power.

About 20% of steel produced is exported for specialized application like offshore oil platform anchoring, mining applications, crane elevator, fishing, automobile, structural infrastructure etc. Production process has state-of-the art technology. The process of steel making starts from iron core taken through mini blast furnace, integrated with electric arc furnace & vacuum refined as required.

Liquid steel is cast through continuous casting machines having closed casting, electro magnetic stirrer & high level of automation to impart consistency. Most of the steel is rolled to wire rods in a sophisticated WIRE ROD MILL having in-built facilities like no-twist rolling, low temperature rolling, TC rings and on-line heat treatment. This mill can roll sizes between 5mm to 28.6mm.

This is the first steel plant in the country to win TPM award from JIPM, Japan

1.2 Industrial Automation

The billets are brought from the steel melting shop and billet conditioning is done on the billets. billet conditioning is done using

1. Automatic billet grinding machine (ABGM)
2. Short blast machine (SBM)
3. Manually (Hand grinding)

Grinding is done to remove the defects on the billets. After grinding, the billets are transferred to the billet charging grate using magnetic crane. The charging grate has fixed and moving beams by the help of which the billet is transferred to the Charging Grate Roller Table (CGRT). The billet from the charging grate is brought to the approach roller table. From the approach roller table the billet enters the walking beam furnace.

The entry into the furnace is controlled by photo cells. 3 photo cells are used. The first photocell stops the billet movement by stopping the approach roller table. The second photocell measures the length of the billet. The third photocell stops the charging roller table and positions the billet inside the furnace. When the front end of the billet reaches the third photocell, the gate of the furnace opens and the billet enters the furnace.

The furnace has rollers, which are called charging roller table. Inside the re-heating furnace the billets are heated to a temperature around 1100 to 1200 degree centigrade. The heated billets then come out of the furnace. At the end of the furnace there are the Discharge roller table and furnace exit table and descenders. After the descenders the billets enter the stands.

29 stands are used here for rolling the billets to form rods. All the stands are driven by D.C. motors. Stand 1 and stand 2 are used for horizontal compression. Stand 3 is used for vertical compression. From stand 4 alternately horizontal and vertical stands are placed. The stands are used to mould the billet from a square cross section to a circular cross section gradually, decreasing its diameter from one stand to another stand. In between some of the stands vertical loppers are used to reduce tension in the metal. Shear 7, shear 13 and shear 19 are used to cut the ends of the billet which cool down.

After stand 19 the heated metal is sent either to the block or to the garret depending upon the diameter. For 5mm - 17.5mm the heated metal are passed to the block and for diameter ranging from 18mm - 28.6mm the heated metal is sent to the garret where they are coiled up.

In the block there are 10 stands, which are operated by 2 D.C. motors. In these stands also diameter is reduced considerably. After the stands there is the cooling section known as water zone. There are 6 zones for cooling. After the cooling section in the block there is a laying head, which is used to coil up the metal.

The coiled metals are then moved through conveyors, which have blowers below from which air is blown to cool down the coils. Then the coils are piled up in the reform tub. From the reform tub the coils are placed in pallets and are moved using conveyors. Proximity switches control the movement of the loaded pallets conveyors. From the pallet conveyors the coils are transferred to the hook conveyor (also known as the power and free conveyor) at the transfer section. Hook conveyors carry the coil to the compacting and binding zone. Then the coils are compressed using a compacter and bound by 4 binding unit. Now the coils are taken out from the hook conveyors with the help of unloaders and taken to the yard through forklifts where the coils are finally ready for dispatch.

The mill is divided into many parts:-From stand 1 to stand 5 it is called the roughing mill part. From stand 6 to stand 13 it is known as the intermediate mill 1. From stand 14 to stand 19 it is called as intermediate mill 2. From intermediate mill 2 till the end of the mill is known as the block.

Modern manufacturing industries rely extensively on automation to achieve high productivity, repeatability, operational safety and equipment availability. Industrial automation involves the coordinated operation of sensors, controllers, actuators, drives, communication systems and operator interfaces.

At the centre of many industrial automation systems is the **Programmable Logic Controller (PLC)**. A PLC continuously receives information from field devices, executes a programmed control strategy and generates outputs that influence the physical process.

The reference material describes the PLC as an industrial computer designed for demanding industrial environments, with important advantages including reliability, fast response and flexibility in program modification.

For automation in the mill, the mill is divided into various sections ,so that they may be controlled easily by various PLC's.

The process includes:

- Billet conditioning
- Billet charging
- Reheating
- Descaling
- Rolling
- Shearing
- Cooling
- Coiling
- Conveyor transportation
- Compacting
- Binding
- Dispatch

The billets are heated in the reheating furnace to approximately **1100–1200°C** before entering the rolling stands. The rolling section uses multiple stands

driven by DC motors.

1.3 Learning-to-Project Progression

The internship provided a logical transition from industrial automation concepts to the developed monitoring prototype.

Industrial Concept	Project Implementation
PLC CPU and program logic	Node-RED function-based processing
Analog/digital I/O	Temperature, vibration and current variables
Industrial communication	MQTT-based data path
HMI/SCADA concept	Node-RED Dashboard
Troubleshooting	Debugging and signal validation
Reliability/redundancy	Consideration for future deployment
Motor control/drives	Motor current and operating status monitoring

This connection allowed the project to move from theoretical industrial automation concepts toward an applied digital monitoring system.

2. PLC FUNDAMENTALS AND SYSTEM ARCHITECTURE

2.1 What is a PLC?

A **Programmable Logic Controller** is an industrial digital controller designed to monitor inputs, execute programmed logic and control outputs.

The basic operating sequence can be represented as:

Input Acquisition → Program Execution → Output Update → Communication/Status

The PLC continuously reads field signals, processes them according to the control program and updates the connected output devices.

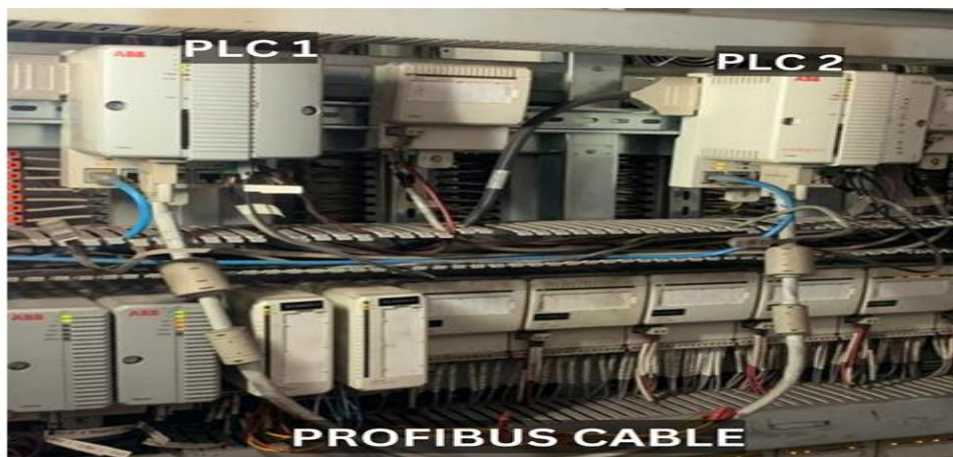
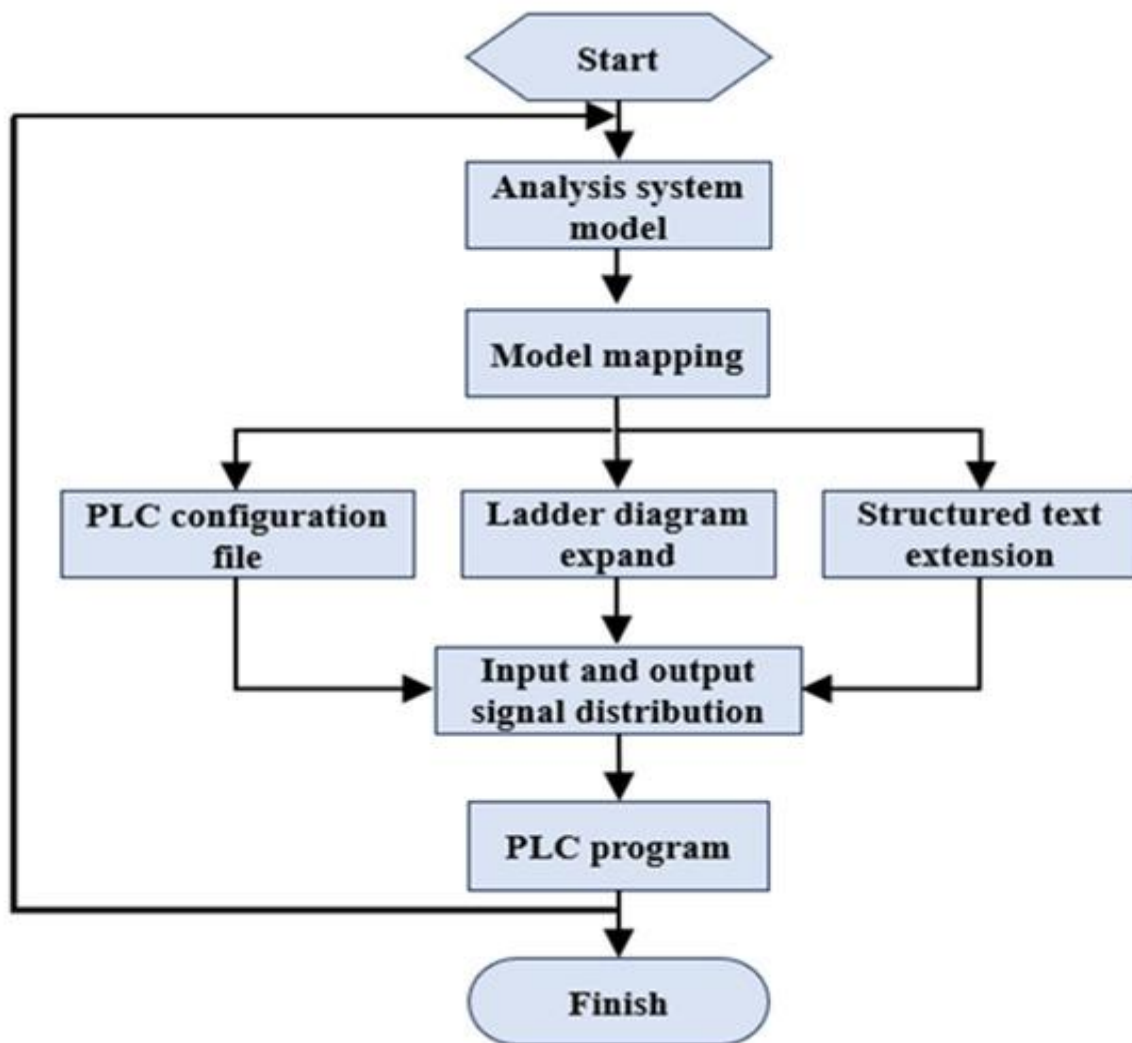


ABB COMPANY'S PLC



SIEMENS COMPANY'S PLC

2.2 PLC Control Logic



2.3 Major Components of a PLC

2.3.1 CPU

The Central Processing Unit is responsible for executing the control program and handling input/output operations.

The CPU of PLCs are connecting using and COMMUNICATION INTERFACE CARD.



ABB'S COMMUNICATION INTERFACE CARD



SIEMENS'S COMMUNICATION INTERFACE CARD

2.3.2 Input Modules

Input modules receive information from field devices.

Examples include:

- Push buttons
- Limit switches
- Proximity sensors
- Temperature transmitters
- Pressure transmitters
- Current measurement devices

2.3.3 Output Modules

Output modules allow the PLC to influence the physical process.

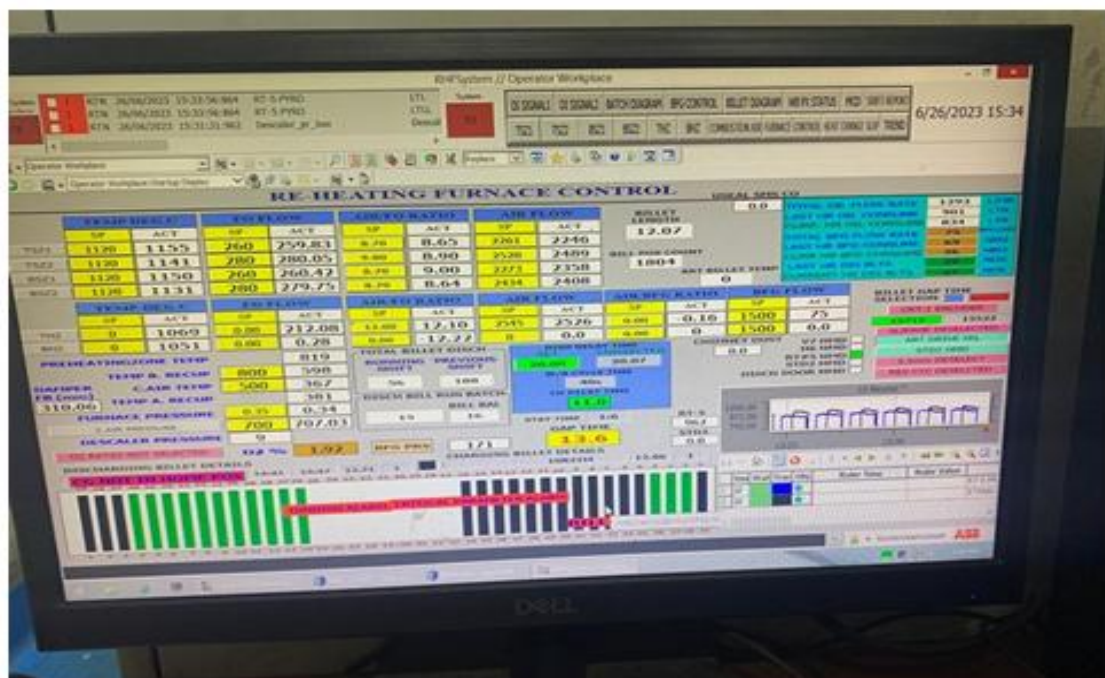
They may operate:

- Relays
- Contactors
- Solenoid valves
- Lamps
- Motors
- Drives

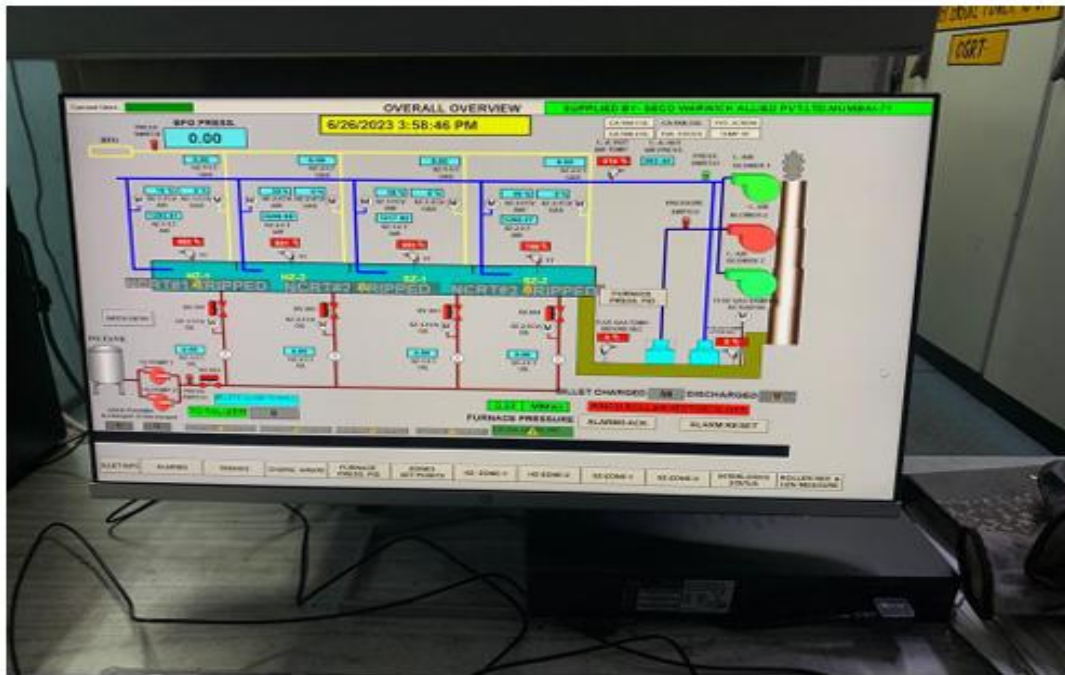
2.3.4 Communication Interface

The communication interface allows the PLC to exchange information with:

- Remote I/O
- HMIs
- SCADA systems
- Other PLCs
- Industrial networks
- Supervisory systems



ABB'S PLC'S HUMAN-MACHINE INTERFACE



SIEMENS'S PLC'S HUMAN-MACHINE INTERFACE

2.3.5 Memory

PLC memory stores the control program and process-related data.

2.3.6 Power Supply

The power supply provides the required electrical power to the PLC system and associated modules.

2.4 PLC and Monitoring Layer

A critical engineering principle in this project is that the monitoring system **does not replace the PLC**.

The PLC continues to perform:

- Deterministic control
- Interlocking
- Process sequencing
- Control decisions
- Approved safety-related functions.

The monitoring layer focuses on:

- Visualization
- Trend analysis
- Alarm context
- Equipment condition
- Maintenance information

This separation makes the proposed architecture more suitable for industrial implementation.

3. PLC INPUTS, OUTPUTS AND FIELD INSTRUMENTATION

3.1 Digital Inputs

Digital inputs represent discrete conditions. Examples include:

- Motor ON/OFF
- Push-button status
- Limit-switch status
- Proximity sensor state
- Alarm status

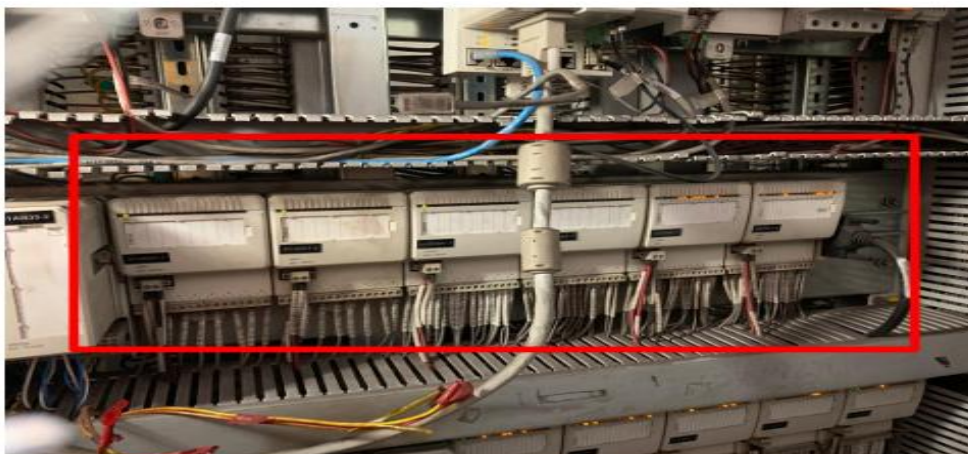
A digital input generally represents two logical states such as ON/OFF or 1/0.

3.2 Analog Inputs

Analog inputs represent continuously varying physical quantities. Examples include:

- Temperature
- Pressure
- Level
- Flow
- Current
- Vibration

For the motor-monitoring application, temperature, vibration and current are treated as continuously varying condition parameters. Motor operating status and alarm state can be represented as digital information.



ANALOG AND DIGITAL INPUT

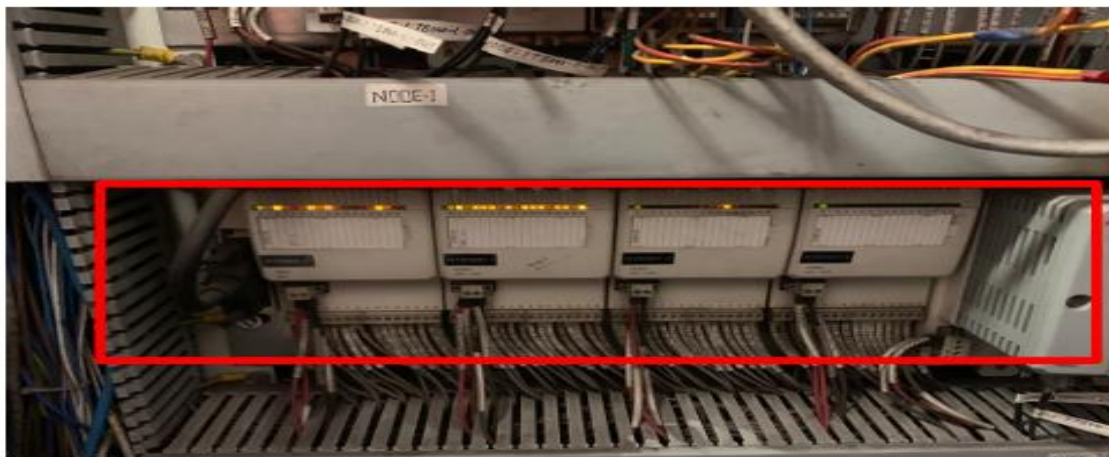
3.3 Digital Outputs

Digital outputs can be used to control discrete devices such as:

- Relays
- Contactors
- Solenoid valves
- **Indicator** lamps

3.4 Analog Outputs

Analog outputs can be used to control continuously variable equipment such as drives and control valves.



ANALOG AND DIGITAL OUTPUT

3.5 to Motor Monitoring

In a real implementation, suitable motor sensors could provide measurements that enter the PLC or condition-monitoring hardware. These values could then be scaled into engineering units before being transferred to the monitoring system.

The prototype follows the same conceptual process:

Physical Parameter → Measurement → Data Value → Processing → Dashboard

4. INDUSTRIAL NETWORK COMMUNICATION AND RELIABILITY

4.1 Importance of Industrial Networking

Modern industrial plants contain many controllers, sensors, remote I/O systems, HMIs and supervisory computers. Communication networks allow these devices to exchange information.

A reliable communication network is important because loss of communication can affect:

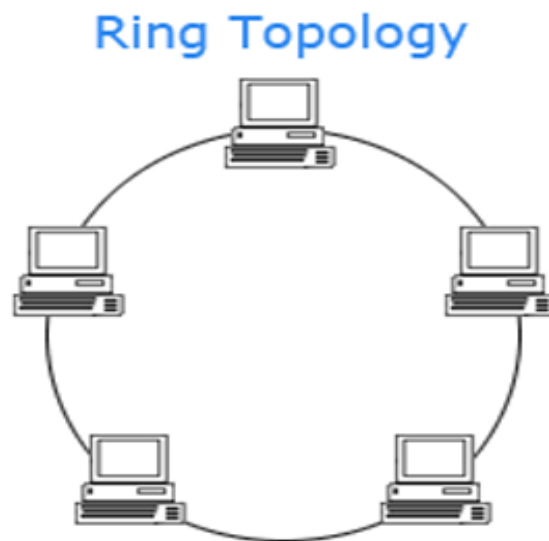
- Monitoring
- Diagnostics
- Process visibility
- Maintenance
- System availability

4.2 Ring Topology

A ring network connects devices in a closed loop.

One major advantage is that appropriately designed redundant ring architectures can provide an alternate communication path.

However, ring networks may require more careful configuration and diagnosis.



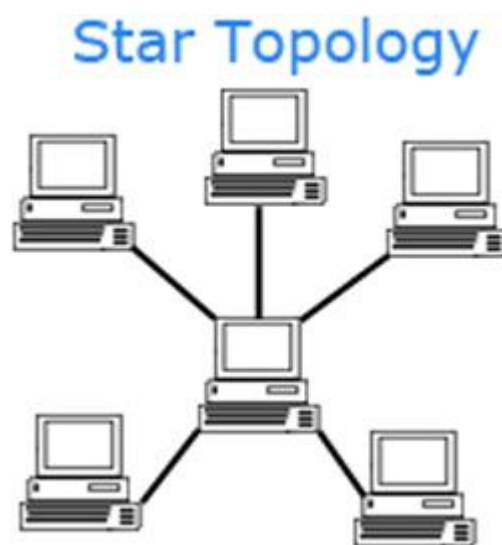
4.3 Star Topology

In a star topology, devices connect to a central switch.

Advantages include:

- Simple expansion
- Easy fault isolation
- Straightforward network organization

However, the central network device becomes an important component whose failure can affect connected devices.



4.4 Industrial Communication Media

Industrial communication may use different physical media depending on distance, speed and environmental requirements.

The reference material includes:

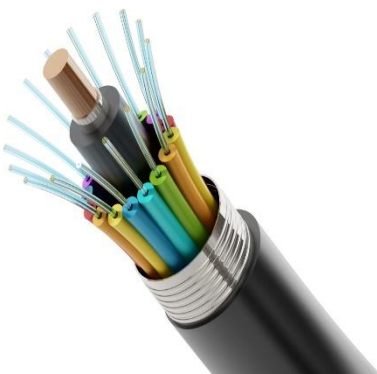
- Ribbon Cables
- Fiber Optic Cables
- Profibus cable

- **Ribbon Cables**



In PLC connections, ribbon cables are typically used for internal connections between the CPU module and other modules within the control rack or cabinet. These modules can include input and output modules, communication modules, power supplies, and other peripheral devices.

- **Fiber Optic Cables**



Fiber optic cables are used in PLC (Programmable Logic Controller) connections to enable high-speed and long-distance communication between various components in the control system. Fiber optic cables transmit data using light signals, offering several advantages over traditional copper cables, such as higher bandwidth, immunity to electromagnetic interference, and longer transmission distances.

- **Profibus cable**



Profibus is a popular fieldbus protocol commonly used in industrial automation systems, including PLC (Programmable Logic Controller) connections. Profibus cables are specifically designed to carry data signals in Profibus networks, facilitating reliable and high-speed communication between PLCs and other devices.

4.5 Redundant Power

A redundant power arrangement uses multiple power sources so that failure of one source does not necessarily stop the system.

A suitable arrangement can provide:

Primary Supply → Failure Detection → Automatic Switchover → Backup Supply → Restoration

This principle is relevant to the monitoring system because the monitoring data path should itself be reliable and should not become a single point of failure for the control system.

5. SIEMENS IM 153-1 INTERFACE AND REMOTE I/O

The **Siemens IM 153-1** is an interface module used in the referenced industrial architecture to connect a PLC CPU with distributed I/O modules. Its role includes communication between the controller and remote I/O equipment.

The reference material associates the interface with:

- CPU-to-I/O communication
- PROFIBUS DP communication
- I/O expansion
- Diagnostic information
- Maintenance-related replacement capability



SIEMENS S7 300 PLC IM-153-1

5.1 Relevance to Motor Monitoring

In an industrial motor-monitoring application, sensors may not necessarily be located close to the PLC CPU.

Remote I/O provides a way to acquire signals at distributed locations.

For example:

Motor Sensors → Remote I/O → PLC → Industrial Network → Monitoring Gateway

A monitoring system can then consume an approved subset of these values without modifying the fundamental PLC control logic.

6. PLC TROUBLESHOOTING METHODOLOGY

6.1 Importance of Troubleshooting

Troubleshooting is an essential part of industrial automation engineering.

A fault should not be diagnosed by randomly replacing components. Instead, the engineer should trace the signal systematically from the source to the destination.

6.2 Troubleshooting Procedure

The following methodology was studied:

Step 1 — Understand the symptom

Identify exactly what is wrong.

Step 2 — Check power

Verify:

- PLC power
- I/O power
- Network power
- Gateway power

Step 3 — Inspect wiring

Check:

- Terminals
- Cables
- Junction boxes
- Field connections

Step 4 — Check the field device

Verify whether the sensor or transmitter is functioning correctly.

Step 5 — Check the I/O

Confirm that the PLC or remote I/O is receiving the expected signal.

Step 6 — Check the signal value

Compare the measured value with the expected engineering range.

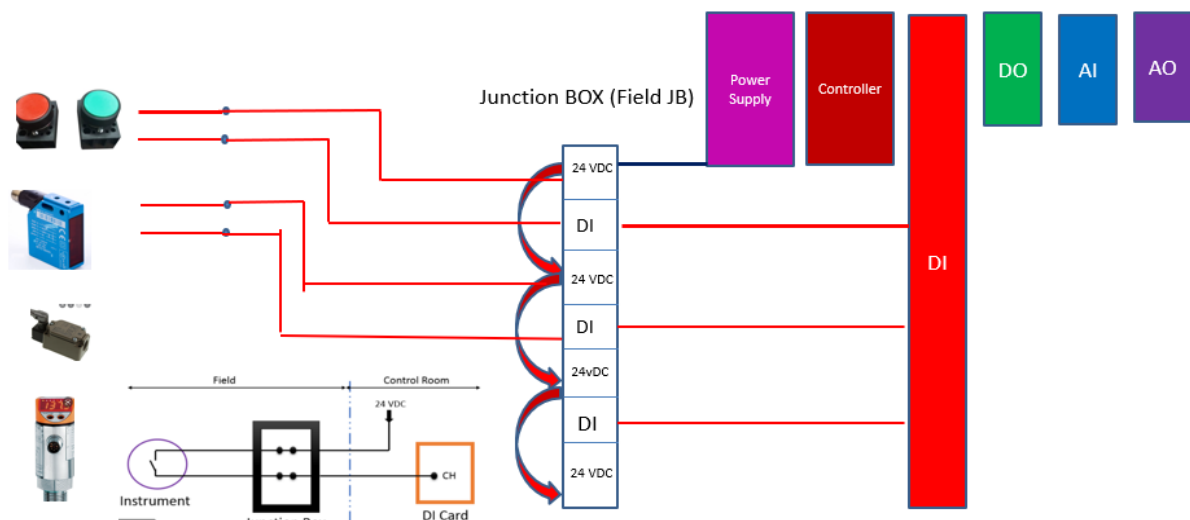
Step 7 — Check the control logic

Review:

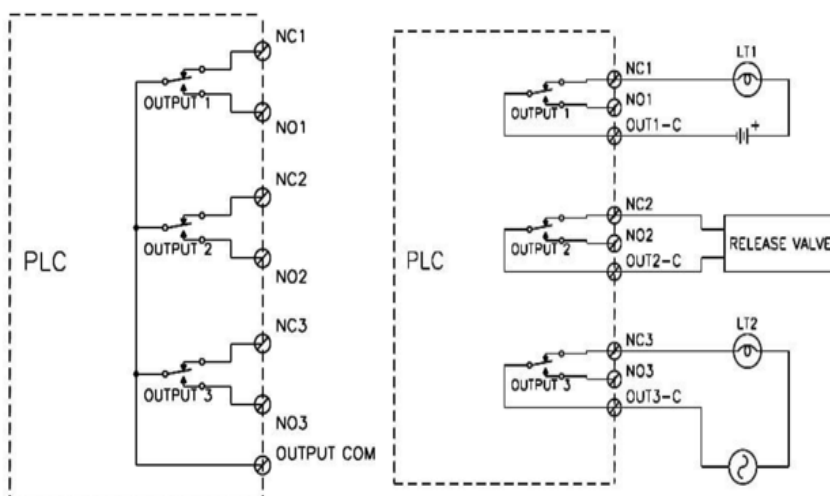
- Program logic
- Interlocks
- Conditions
- Sequencing

Step 8 — Correct and verify

After corrective action, confirm that the system operates normally.



PLC Digital Input trouble shooting technique



- Relay output are normally used to control moderate loads (Up to 2 amp or when a very low on resistance is required)
- Always Provide surge suppressor for devices such as solenoids etc .

6.3 Application to the Developed Prototype

The same methodology was directly applied during Node-RED development.

Problem	Investigation
Dashboard chart blank	Message → payload → topic → timestamp → chart configuration
Gauge not updating	Output connection → payload type → numeric value
Motor status incorrect	Source status → function logic → dashboard input
Alarm count incorrect	Alarm condition → context value → message flow
MQTT data unavailable	Broker → topic → publisher → subscriber → debug

This became one of the most useful practical lessons of the project: **trace the signal from source to destination before modifying the system.**

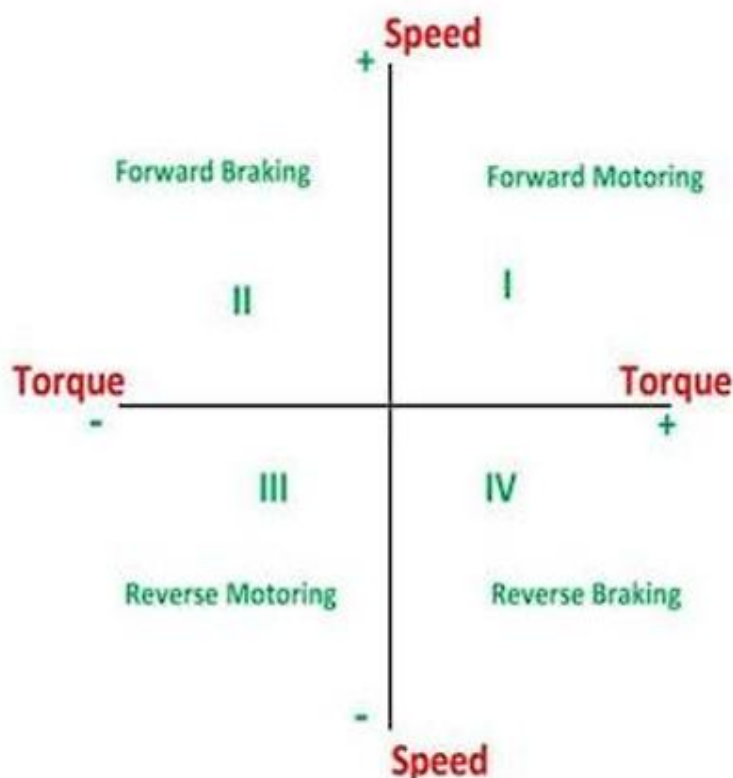
regenerative braking. The armature current is positive, but the voltage is negative as energy is fed back into the drive system.

3. Third Quadrant (Reverse Motoring): In the third quadrant, the motor operates in the reverse direction (counterclockwise) while running as a motor. The armature current is negative, and the voltage is also negative.

4. Fourth Quadrant (Forward Regenerative Braking): In the fourth quadrant, the motor rotates in the reverse direction (counterclockwise), and it acts as a generator, regeneratively braking the load. Both the armature current and voltage are negative as energy is returned to the drive system.

Quadrant Operating Condition

I	Forward motoring
II	Forward rotation with regenerative braking
III	Reverse motoring
IV	Reverse rotation with regenerative braking



7.2 Drive Data for Monitoring

Although the developed prototype is not a drive-control system, drive-related information can become valuable monitoring information.

Future versions could include:

- Motor speed
- Torque
- Drive fault codes
- Operating mode
- Run hours
- Drive temperature

Motor current and operating status were already included as monitoring variables in the prototype.

The engineering boundary is important: the prototype **monitors motor condition but does not issue unsafe motor-control commands**. Actual control must remain under the approved PLC, interlocking and safety architecture.

8. PROBLEM DEFINITION AND PROJECT OBJECTIVES

8.1 Problem Statement

Industrial motors may experience abnormal operating conditions before a major failure becomes obvious.

Important indicators can include:

- Increasing temperature
- Excessive vibration
- Abnormal current
- Repeated alarm conditions

Periodic inspection may not capture rapidly changing conditions.

Therefore, a centralized monitoring layer can provide continuous visibility into motor condition and support maintenance decisions.

8.2 Project Objectives

The major objectives were:

1. Develop a real-time motor-monitoring workflow using MQTT and Node-RED.
2. Monitor temperature, vibration and current.
3. Display motor ON/OFF status.
4. Display alarm status.
5. Calculate a rule-based motor health score.
6. Generate parameter-specific alerts.
7. Maintain a running alarm count.
8. Display the latest update time.
9. Generate live trend charts.
10. Develop a centralized monitoring dashboard.
11. Establish a conceptual mapping to PLC/industrial I/O architecture.

8.3 Scope

The implemented project is a **monitoring prototype**.

The current implementation uses MQTT-generated motor data rather than a direct live PLC connection.

The following are considered future extensions:

- Direct PLC integration
- Historian/database storage
- Machine-learning prediction
- Industrial cybersecurity implementation
- High-availability deployment

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9. PROPOSED MOTOR MONITORING ARCHITECTURE

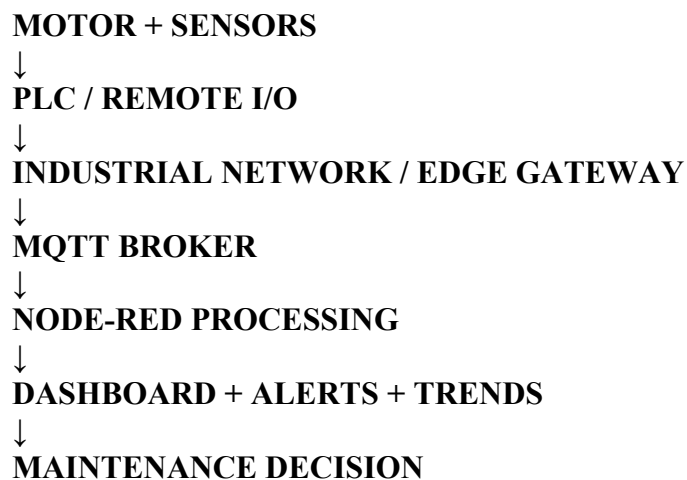
9.1 Overall Architecture

The proposed system consists of multiple layers.

Layer	Function	Implementation
Field Layer	Measure motor condition	Temperature, vibration, current sensors
Control Layer	Deterministic control	PLC + I/O
Communication Layer	Transfer selected data	Industrial network/gateway/MQTT
Processing Layer	Process and classify data	Node-RED
Visualization Layer	Present information	Node-RED Dashboard
Maintenance Layer	Support decisions	Inspection and planned maintenance

9.2 End-to-End Data Flow

The proposed architecture is:



9.3 Engineering Principle

The architecture separates control from monitoring. The control system remains responsible for safe and deterministic operation. The monitoring layer consumes selected data and transforms it into:

- Equipment visibility
- Trend information
- Alarm information
- Health indication & Maintenance intelligence

10. MQTT AND NODE-RED IMPLEMENTATION

10.1 MQTT Communication

MQTT was used as the data-entry mechanism for the prototype.

The implemented MQTT topic is:

Industrial/motor

The MQTT input node receives the motor data and sends it to the Node-RED function node.

10.2 Node-RED Processing

The Node-RED function node performs the main processing operations.

The sequence is:

Receive Data → Extract Values → Evaluate Conditions → Calculate Health → Generate Alerts → Count Alarms → Timestamp → Distribute Outputs

The function extracts:

- Temperature
- Vibration
- Current
- Motor status
- Alarm status

It then generates separate outputs for the dashboard components.

10.3 Dashboard Outputs

The developed flow includes:

- Temperature Gauge
- Vibration Gauge
- Current Gauge
- Motor Status
- Alarm Status
- Temperature Trend
- Vibration Trend

- Current Trend
- Health Score
- Temperature Alert
- Vibration Alert
- Current Alert
- Total Alarms
- Last Update

Figure 10.1 — Developed Node-RED Motor Monitoring Flow

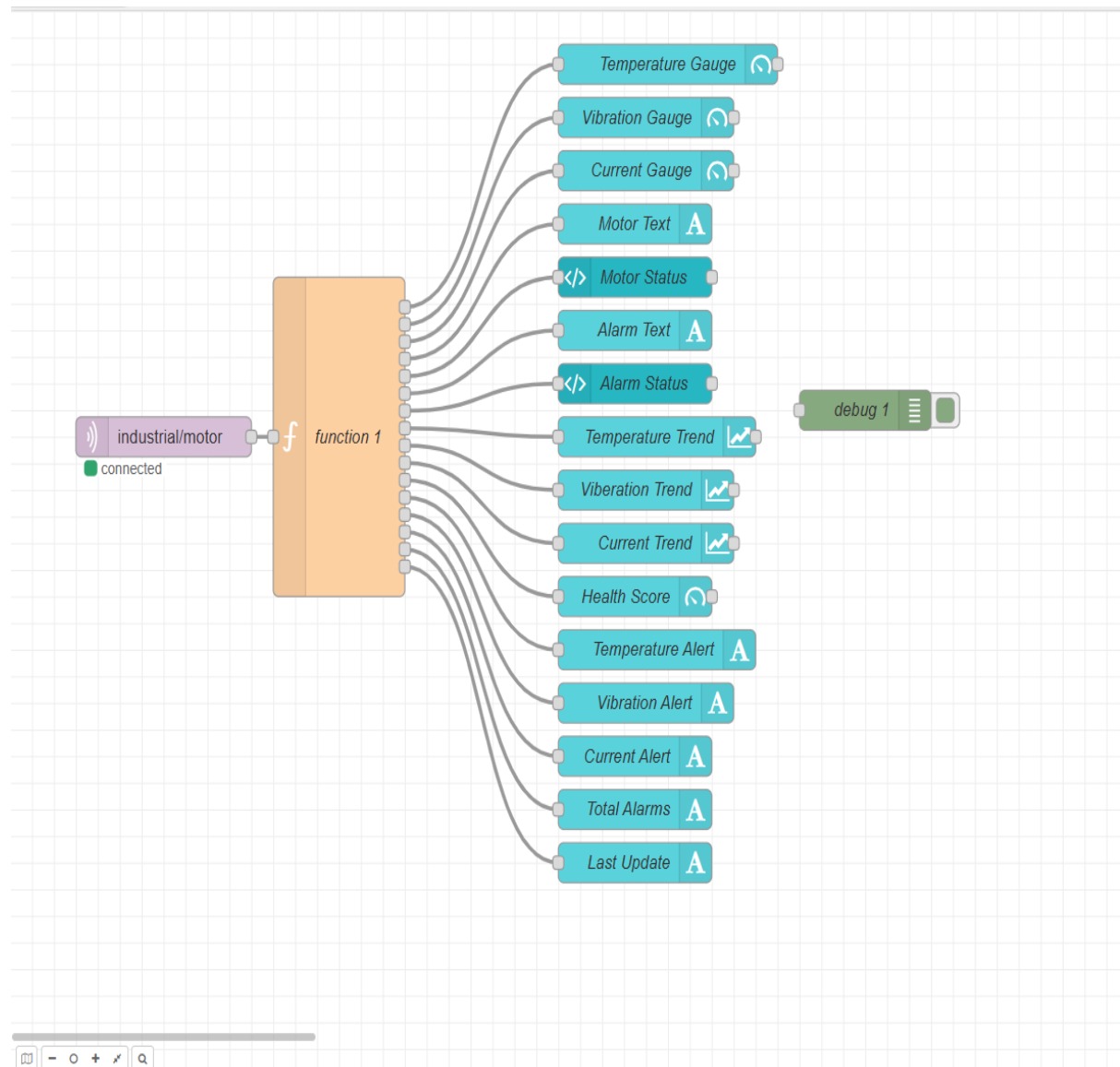


Figure 10.1: Node-RED motor-monitoring flow showing MQTT input, function-based processing and dashboard outputs.

11. MOTOR HEALTH SCORE AND ALARM LOGIC

11.1 Health Score

The prototype uses a **transparent rule-based health score** rather than a machine-learning model.

The initial health score is:

$$\text{Health} = 100 - (0.4 \times \text{Temperature}) - (3 \times \text{Vibration}) - (2 \times \text{Current})$$

The resulting value is constrained to the practical range:

$$0 \leq \text{Health Score} \leq 100$$

The score is intended as a **prototype condition index** and should not be interpreted as a statistically validated probability of failure.

11.2 Alert Classification

The implemented thresholds are:

Parameter	Normal	Medium / Warning	High
Temperature	$\leq 50^{\circ}\text{C}$	$>50^{\circ}\text{C}$	$>70^{\circ}\text{C}$
Vibration	$\leq 4 \text{ mm/s}$	$>4 \text{ mm/s}$	$>7 \text{ mm/s}$
Current	$\leq 10 \text{ A}$	$>10 \text{ A}$	$>15 \text{ A}$

These values were used for prototype demonstration.

In an actual industrial plant, thresholds must be selected based on:

- Equipment ratings
- Manufacturer recommendations
- Operating conditions
- Historical data
- Approved maintenance standards

11.3 Alarm Counter

The Node-RED flow uses context storage to maintain the total number of alarms.

When the incoming alarm state is **ON**, the alarm counter increases. This provides the operator with a simple indication of the number of alarm events observed by the monitoring application.

12. DASHBOARD DESIGN AND HUMAN-MACHINE INTERACTION

12.1 Dashboard Philosophy

The dashboard was designed to provide an operator-oriented view rather than simply displaying unrelated widgets.

The information hierarchy was defined as:

Current Condition → Status → Trends → Maintenance Context

The dashboard contains the following major areas:

Live Gauges

Displays:

- Temperature
- Vibration
- Current
- Health Score

Motor Information

Displays:

- Motor ON/OFF
- Alarm ON/OFF
- Alarm count

Live Trends

Displays:

- Temperature trend
- Vibration trend
- Current trend

Alert Information

Displays the condition classification of the monitored parameters.

Last Update

Shows the latest data-update timestamp.

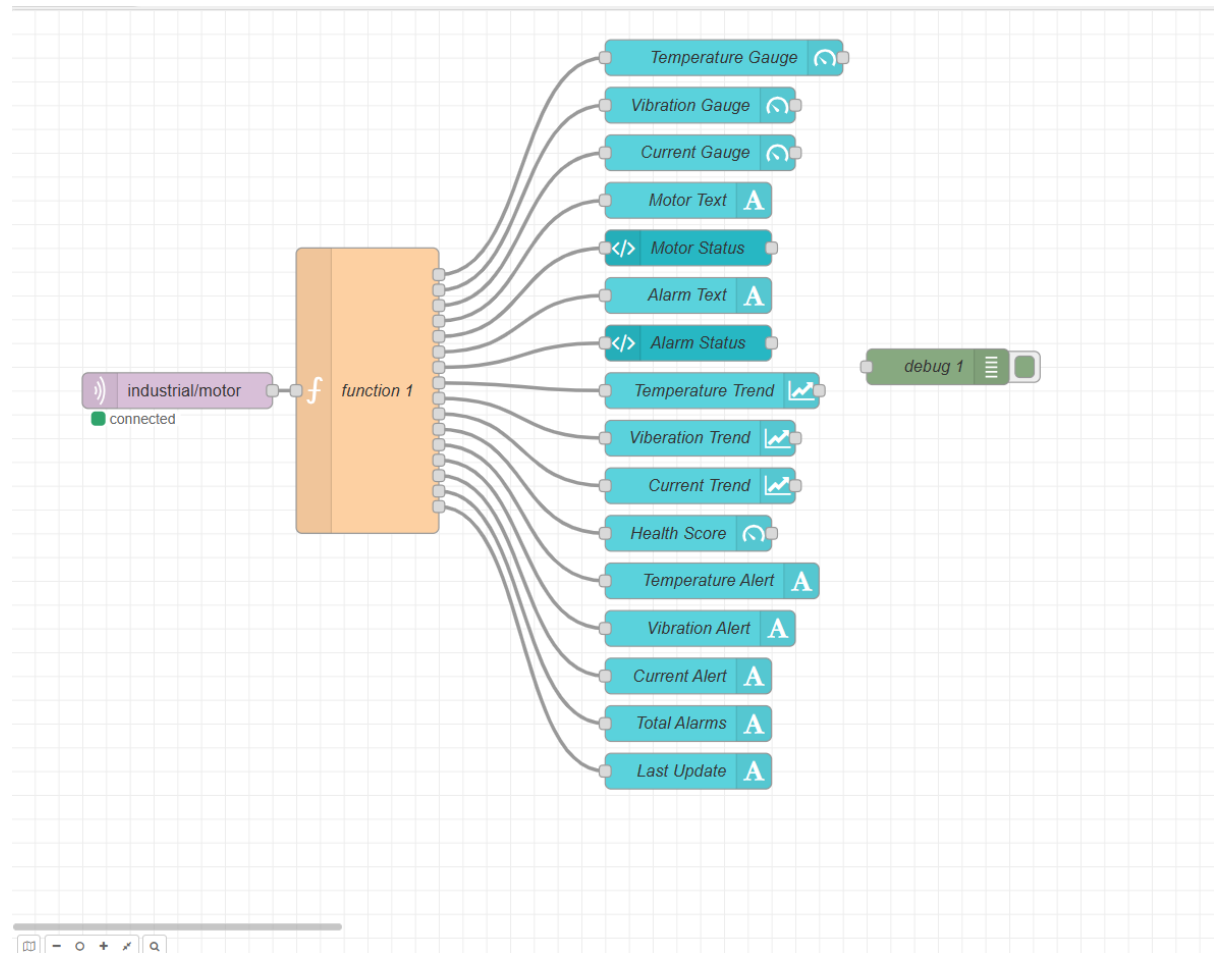


Figure 12.1 — Final Motor Monitoring Dashboard

Figure 12.1: Developed Node-RED Dashboard showing live gauges, motor status, alarm status, health score and real-time parameter trends.

12.2 Dashboard Development and Debugging

During development, the gauges were successfully receiving data while the charts initially appeared blank.

The issue was traced to the message structure expected by the chart nodes.

The final implementation assigns:

- Numeric payload values
- Appropriate topic labels
- Timestamp information

This allows the chart widgets to plot temperature, vibration and current correctly.

This debugging process demonstrated a strong connection between software troubleshooting and PLC troubleshooting: **trace the signal, verify the value, verify the destination configuration and then modify the system if necessary.**

13. TESTING, RESULTS AND OBSERVATIONS

13.1 Functional Testing

The prototype was tested under different simulated motor conditions.

Test Condition	Expected Behaviour	Observed Behaviour
Normal temperature	Normal temperature status	Normal condition displayed
Temperature above warning	Warning classification	Warning displayed
Temperature above high threshold	High alert	High condition displayed
Normal vibration	Normal condition	Gauge/trend updated
High vibration	Vibration alert	Alert classification changed
Normal current	Normal condition	Gauge/trend updated
High current	Current alert	Alert classification changed
Motor ON	ON status	Dashboard displayed ON
Motor OFF	OFF status	Dashboard displayed OFF
Alarm ON	Counter increases	Total alarm count increased
Continuous data	Trends update	Charts displayed historical values

13.2 Results

The final prototype successfully demonstrated:

- Real-time parameter display
- Temperature monitoring
- Vibration monitoring
- Current monitoring
- Motor status monitoring
- Alarm status monitoring
- Health-score calculation
- Threshold-based alerts
- Alarm counting

- Timestamp display
- Historical trend visualization
- Centralized dashboard monitoring



Figure 13.1 — Final Working Dashboard

14. INDUSTRIAL DEPLOYMENT CONCEPT

14.1 Connection to a Real Industrial Plant

The developed prototype can be considered a supervisory layer that could be integrated into an industrial environment through an approved architecture.

The proposed deployment would **not directly connect an experimental Node-RED system to critical control outputs**.

Instead, the existing PLC/control system would continue performing its approved control functions.

A suitable deployment sequence would be:

1. Install suitable temperature, vibration and current sensors.
2. Connect the sensors to appropriate PLC analog/digital I/O or approved monitoring equipment.
3. Validate signal scaling.
4. Configure appropriate alarm limits.
5. Expose selected monitoring tags through an approved gateway.
6. Transfer the required data to the monitoring broker/historian.
7. Process the data using Node-RED.
8. Display the information on the dashboard.
9. Use trends and alerts to support inspection and maintenance decisions.

14.2 Potential Industrial Applications

The architecture could potentially be applied to:

- Conveyor drive motors
- Pump motors
- Fans and blowers
- Compressor auxiliary motors
- Cooling-system motors
- Rolling/mill auxiliary equipment

14.3 Industrial Value

Condition Signal → Abnormality Detection → Maintenance Attention → Corrective Action → Verification

This provides a conceptual foundation for condition-based and eventually predictive maintenance.

15. RELIABILITY, MAINTENANCE AND CYBER/OPERATIONAL CONSIDERATIONS

15.1 Reliability

A monitoring system is useful only when its own data path is sufficiently reliable. A production implementation should therefore consider:

- Reliable power
- Redundant or highly available systems
- Network resilience
- MQTT broker availability
- Local data buffering
- Recovery after restart

15.2 Deployment Troubleshooting Workflow

A structured industrial troubleshooting workflow can be represented as:

Step	Question
1. Symptom	What exactly is wrong?
2. Power	Are PLC, I/O and network devices powered?
3. Field	Are sensors and wiring healthy?
4. I/O	Does the PLC/remote I/O receive the expected signal?
5. Network	Is the communication path working?
6. Data	Is the value reaching the gateway/MQTT layer?
7. Application	Is Node-RED interpreting the data correctly?
8. Action	Has the problem been corrected and verified?

15.3 Safety Boundary

The monitoring dashboard must not be treated as a safety system. Safety functions such as:

- Emergency stops
- Safety interlocks
- Protection systems
- Critical control functions

must remain within the approved industrial control and safety architecture. The dashboard is intended as a **decision-support and supervisory interface**.

16. LIMITATIONS AND FUTURE SCOPE

16.1 Current Limitations

The current prototype has several limitations:

1. It uses MQTT motor data rather than direct communication with a live industrial PLC.
2. The health score is rule-based.
3. The health score has not been statistically validated against historical failures.
4. Long-term historian/database storage is not currently implemented.
5. Machine-learning-based failure prediction is not implemented.
6. Industrial cybersecurity, redundancy and network segmentation require formal engineering before deployment.

These limitations are important because they define the boundary between a **working prototype** and a fully commissioned industrial predictive-maintenance system.

16.2 Future Scope

The project can be extended through:

1. Real PLC Integration

Connect actual PLC tags through an approved industrial gateway.

2. Historical Database

Store temperature, vibration and current data for long-term analysis.

3. Motor Run-Hour Tracking

Track operating hours to support maintenance scheduling.

4. Drive Information

Add:

- Speed
- Torque
- Fault codes
- Operating mode

5. Machine Learning

Use labelled historical failure data to develop anomaly-detection and failure-prediction models.

6. Remaining Useful Life

With sufficient historical data, the system could potentially estimate Remaining Useful Life (RUL).

7. Maintenance Recommendations

The system could evolve from simply generating alarms to suggesting appropriate maintenance actions.

8. Cybersecurity

Future deployment could include:

- Role-based access
- Authentication
- Audit logging
- Secure communication
- Network segmentation

9. High Availability

Redundant gateway and broker architectures could be considered for critical applications.

16.3 Evolution Toward Predictive Maintenance

The proposed development path is:

Rule-Based Monitoring

↓

Historical Data Collection

↓

Feature Engineering

↓

ML-Based Anomaly Detection



Failure Prediction / RUL



Predictive Maintenance

17. CONCLUSION

This internship project provided an opportunity to study industrial automation from both a theoretical and practical perspective.

The first part of the project established an understanding of **PLC architecture, CPU and I/O modules, communication interfaces, HMI concepts, industrial networking, redundant power systems, remote I/O and troubleshooting methodology.**

The study of industrial communication introduced concepts such as ring and star network topologies, industrial communication media and Siemens IM 153-1 remote-I/O architecture. The troubleshooting study further developed a structured method for identifying problems related to power, wiring, I/O, communication and signal processing.

The second part of the project transformed these concepts into a practical **motor condition-monitoring prototype using MQTT and Node-RED.**

The developed system receives motor-related data, processes temperature, vibration and current values, evaluates predefined thresholds and produces a rule-based health score. The information is then presented through a centralized dashboard containing gauges, motor status, alarm status, health score, alerts, trend charts, alarm count and update information.

The most important engineering outcome of the project is the connection between **industrial control and maintenance intelligence.**

The PLC remains responsible for reliable and deterministic control, while the monitoring layer provides additional visibility into equipment condition. This separation allows the monitoring system to support maintenance decisions without replacing the approved control and safety architecture.

With future integration of real PLC data, historical databases, validated condition indicators and machine-learning techniques, the prototype can be developed further into a more advanced predictive-maintenance platform.

Ultimately, the project demonstrates an end-to-end engineering approach:

Understand the Plant → Understand the PLC → Understand the Signal Path → Build the Monitoring Layer → Test and Troubleshoot → Support Maintenance → Move Toward Predictive Maintenance.

18. REFERENCES AND APPENDIX

18.1 Reference Materials

The following materials were used as the technical foundation of the report:

Reference 1

PLC & Networking

Used for:

- PLC fundamentals
- PLC components
- Redundant power
- Ring/star networks
- Industrial communication
- Siemens IM 153-1
- Troubleshooting
- DC-drive concepts

Reference 2

Basic Procedure for Troubleshooting in PLC

Used for:

- PLC architecture
- Digital I/O troubleshooting
- Relay outputs
- Analog transmitter connections
- PLC/HMI/drive examples

Reference 3

Developed Node-RED Motor Monitoring Prototype

Used for:

- MQTT input
- Node-RED processing
- Health-score calculation
- Alarm logic
- Dashboard gauges
- Trend charts
- Motor/alarm status
- Dashboard design